



Analysis of reasons for failure to return to sport after Latarjet surgery: a systematic review

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Background: The purpose of this study was to perform a systematic review of the literature to identify Shoulder Instability Return to Sport after Injury (SIRSI) scores in athletes who underwent open Latarjet surgery, determine the reasons why athletes failed to return to play (RTP) after Latarjet surgery, and compare SIRSI scores of those who did vs. did not RTP.

Methods: According to PRISMA guidelines, the PubMed, SportDiscus, and Ovid MEDLINE databases were queried to identify studies evaluating return to sport after Latarjet surgery. Study quality was assessed using the MINORS criteria. Studies were included if RTP after Latarjet surgery and a psychological factor were evaluated, with potential psychological factors including readiness to RTP and reasons for failure to RTP.

Results: Fourteen studies, 10 of level III evidence and 4 of level IV evidence, with 1034 patients were included. A total of 978 athletes were eligible to RTP. Of these, 792 (79%) successfully returned to play and 447 (56.4%) returned to play at their previous level of play. Mean RTP time was 6.2 months. Postoperative SIRSI scores averaged 71.2 ± 8.8 at a mean of 21 months' follow-up. Postoperative SIRSI scores for those able to RTP was 73.2, whereas athletes unable to RTP scored an average of 41.5. Mean postoperative SIRSI scores for those in contact sports was 71.4, whereas those in noncontact sports was 86.5. There were 31 athletes with a documented reason why they did not RTP, with postoperative shoulder injury being the most common reason (54.5%). Fear of reinjury and feeling "not psychologically confident" each represented 6.5% of the total.

Conclusion: Athletes who RTP have higher average SIRSI scores than those who are unable to RTP. Of the athletes who documented why they did not RTP, more than half cited a shoulder injury as their reason for not returning to play, whereas fear of reinjury and lack of psychological readiness were other common reasons.

Level of evidence: Level IV; Systematic Review

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Keywords: SIRSI score; Latarjet; psychological readiness; return to play; return to sport; shoulder

Institutional review board approval was not required for this study as this work was derived from anonymized data.

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Anterior glenohumeral instability is a common injury seen in sports that can limit athletes from playing at their current level of competition and ultimately sideline them from the sport altogether.^{21,31} Open transfer of the coracoid process with the attached short head of the biceps and

coracobrachialis to the anterior glenoid, commonly known as the Latarjet procedure, has been shown to be effective and efficient in treating anterior glenohumeral instability, with or without bone loss, returning athletes to their sport up to 97% of the time in reported studies.^{5,10,19,24} Previous studies have outlined the physical factors that are important in determining an athlete's ability to return to play (RTP).^{1,16,20} More recently, however, psychological factors have been shown to be key components in recovery and RTP in the injured athlete.^{6,7,11,23,31}

The Anterior Cruciate Ligament–Return to Sport after Injury was developed by Webster et al in 2008,³³ and was later converted into the Shoulder Instability–Return to Sport after Injury (SIRSI) by Gerometta et al in 2018.^{4,12} The 12-question survey was deemed valid and reproducible to identify patients to return to the same sport after injury that were treated surgically or nonsurgically (Supplementary Appendix S1).¹² The scale itself is a Likert scale that ranges in options from 0 to 10. The 12-question survey is then added up and the overall score is multiplied by 100 before being divided by 12. This final number can then be considered as a percentage.¹² The SIRSI score has now been used to determine readiness for those athletes who underwent Latarjet both preoperatively and postoperatively.^{3,8,9,14,17-19,28} However, the magnitude of the correlation that SIRSI scores have on RTP after Latarjet is not known.

Therefore, the purpose of this study was to perform a systematic review of the literature to identify SIRSI scores in athletes who underwent open Latarjet surgery, determine the proportion of those who returned to play after Latarjet surgery and compare SIRSI scores of those who did not RTP against those who did RTP. The authors hypothesized that SIRSI scores in those athletes who underwent Latarjet and returned to play will be higher than those who did not RTP.

Methods

This was a systematic review performed according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines with no Institutional Review Board approval or funding required.²² The PubMed, SportDiscus, and Ovid MEDLINE databases were queried in January 2023 to identify original research studies dating from inception to December 31, 2022, evaluating return to sport after Latarjet surgery. Search terms used included (labral repair return play) OR (labral repair return sport) OR (Bankart repair return play) OR (Bankart repair return sport) OR (SLAP repair return play) OR (SLAP repair return sport) OR (Shoulder Instability–Return to Sport after Injury) OR (SIRSI). Two blinded independent reviewers were used to examine study eligibility.

Inclusion and exclusion criteria

Studies were included if RTP after Latarjet surgery was evaluated and a psychological factor was also evaluated, with potential

psychological factors including readiness to RTP, kinesiophobia, and reasons for failure to RTP. Articles were excluded if they were review articles or case reports, did not isolate outcomes for Latarjet surgery patients, did not evaluate RTP, or did not evaluate a psychological factor. Studies that evaluate arthroscopic Latarjet surgery without reporting outcomes specific to open Latarjet surgery were also excluded to maximize homogeneity of the patient populations.

Assessment of study quality

Included nonrandomized studies were evaluated for study quality using the Methodological Index for Non-Randomized Studies (MINORS) score.²⁹ Studies were scored for each question as a 0 (not reported), 1 (reported but inadequate), or 2 (reported and adequate). Noncomparative studies included 8 questions for the MINORS score resulting in a total potential score ranging from 0 (poor study quality) to 16 (high study quality), whereas comparative studies instead included 12 questions with a total potential score ranging from 0 to 24.

Data collection and abstraction

Demographic variables collected from the included studies include level of evidence, patient age and sex, follow-up duration, percentage of glenoid bone loss, and frequency of engaging Hill-Sachs lesions. Postoperative outcomes of interest collected include RTP rate, RTP at previous level of play rate, time until RTP, reasons why athletes did not RTP, and postoperative SIRSI scores. Data were collected for full patient cohorts and also by isolating contact vs. noncontact athletes.

Statistical analysis

Postoperative SIRSI scores were also pooled across relevant studies with long-term SIRSI scores due to similarities in follow-up duration. Statistical significance was set at $P < .05$. All statistical analysis was performed with R studio software (version 3.6.3, Vienna, Austria).

Results

Study characteristics

In total, 14 studies were included in our final analysis (Fig. 1).^{2,3,5,8-10,14,15,17-20,24,27} Of these 14 studies, 10 were level III evidence (71.4%)^{2,5,8,9,14,15,17-19,24} and 4 had level IV evidence (28.6%).^{3,10,20,27} The average MINORS criteria score for the level IV studies (noncomparative) was 10.6 (range, 9–12). Average MINOR scores for the level III studies (comparative) was 17.3 (range, 15–19) (Table I).

Demographics

A total of 1034 patients were included across all studies that underwent open Latarjet surgery (Table II). Of these, the

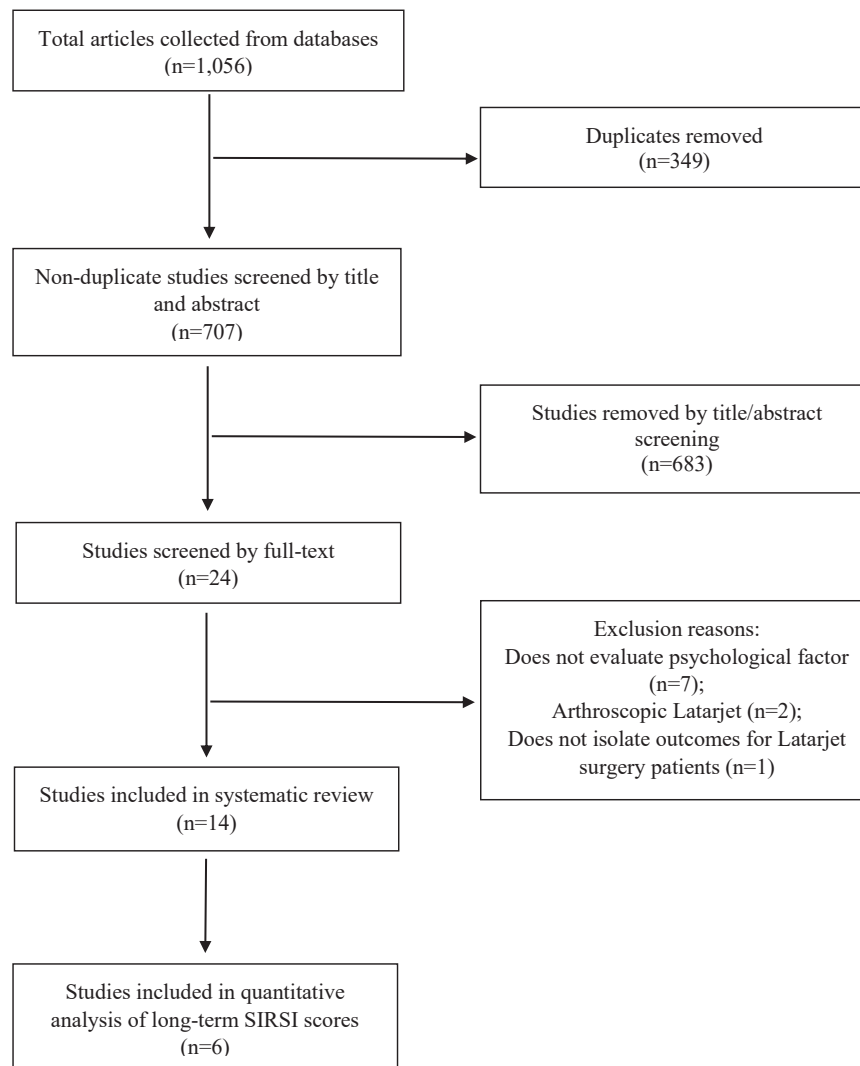


Figure 1 Article selection flowchart. *SIRSI*, Shoulder Instability Return to Sport after Injury.

mean age of the patient population was 26.6 (range, 15-62). Mean follow-up post Latarjet surgery was 3.7 years (standard deviation [SD] 1.28, range 1.1-9.3). Mean glenoid bone loss in the studies that reported ($n = 9$)^{9,10,14,15,17-19,24,27} was 15.5% (SD 3.1%, range 0%-39.1%). Sport classification varied throughout each study (contact sport, collision sport, noncollision sport, overhead, nonoverhead, Gaelic football, hurling, NFL, hockey, rugby). The level of competition ranged from amateur to professional athletes.

SIRSI score

Eight studies included SIRSI scores. Preoperative SIRSI scores had an average of 51.³ Five studies reported SIRSI scores in the postoperative period and had an average score of 71.2 (SD 8.8).^{2,8,10,14,24} When stratifying these postoperative scores into those who did RTP and those who did not RTP, the averages were respectively 73.2 (SD 1.1) and

41.5 (Table III).¹⁷⁻¹⁹ Three studies did not give a raw SIRSI score but did provide an average of 72.2% of athletes scoring >55% on the survey. SIRSI score distribution across reported studies can be seen in Fig. 2. All scores presented averaged greater than the >55% noted passing rate.

RTP

A total of 978 athletes were eligible to RTP. Of these, 792 (79%) successfully returned to play. Of those athletes who successfully returned to play, 447 (56.4%) were noted to have returned to their same level of play. Mean RTP time was 6.2 months (SD 0.9, range 5.1-8.1). Thirty-one athletes who did not RTP documented a reason why. The most common answer was “shoulder injury,” with 17 (54.5%) athletes citing it as their reason (Table IV). Fig. 3 shows the SIRSI score distribution for those who returned to play.

Table I Included studies' characteristics

Year	First author	Journal	Level of evidence	SIRSI score	MINORS score
2022	Castricini ⁵	<i>Arthroscopy</i>	III	N	12
2021	Hurley ¹⁸	<i>Orthop J Sports Med</i>	III	Y	19
2021	Hurley ¹⁹	<i>Orthop J Sports Med</i>	III	Y	19
2022	Paul ²⁴	<i>Arthrosc Sports Med Rehabil</i>	III	N	10
2022	Hurley ¹⁵	<i>Arthrosc Sports Med Rehabil</i>	III	Y	15
2016	Blonna ²	<i>J Shoulder Elbow Surg</i>	III	N	17
2022	Davey ⁹	<i>Arthroscopy</i>	III	Y	17
2022	Ernat ¹⁰	<i>J Orthop</i>	IV	N	9
2020	Rossi ²⁷	<i>Arthroscopy</i>	IV	N	11
2021	Hurley ¹⁴	<i>Arthrosc Sports Med Rehabil</i>	III	Y	19
2021	Davey ⁸	<i>JSES Int</i>	III	Y	19
2022	Hurley ¹⁷	<i>Orthop J Sports Med</i>	III	Y	15
2021	Bohu ³	<i>KSSTA</i>	IV	Y	16
2019	Kordasiewicz ²⁰	<i>KSSTA</i>	IV	N	11

SIRSI, Shoulder Instability–Return to Sport after Injury; MINORS, Methodological Index for Non-Randomized Studies.

Table II Clinical and demographic characteristics of included patients (N = 1034)

Characteristic	Value
Age, mean $\pm$ SD (range)	26.6 $\pm$ 3.3 (15-62)
Male sex, n (%)	934 (90.4)
Female sex, n (%)	100 (9.6)
Mean follow-up, yr, mean (range)	3.9 (3.4-4.5)
Mean glenoid bone loss, % (range)	15.5 (13.3-17.8)
Off-track/engaging Hill-Sachs lesions, n (%)	282 (59.2)
On-track/nonengaging Hill-Sachs lesions, n (%)	90 (27.4)
Recurrent dislocation, % (range)	3.2 (1.8-5.7)
Revision, % (range)	2.6 (1.2-5.4)

SD, standard deviation.

Two (6.5%) athletes did not RTP because of fear of reinjury. Two (6.5%) athletes did not RTP because they believed they were “not psychologically confident.” Five (16.1%) athletes did not RTP because of lifestyle issues whereas another 5 listed their reason as “other.”

Contact vs. noncontact RTP

When stratified by contact and noncontact sport, 83.3% of those in contact sport RTP (190 of 242) and 73.4% returned to their same level of play. Mean RTP was 5.5 months (SD 5.5). Preoperative SIRSI scores had a mean of 51.5 whereas postoperative SIRSI scores had a mean of 71.4 (SD 2.3). Average follow-up duration for postoperative SIRSI score collection was 6.5 months. Those participating in noncontact sports had an RTP rate of 69.2% (45 of 65 reported)

Table III Summary of long-term SIRSI scores

Score	Value
Preop. SIRSI score total, mean	51
Postop. SIRSI score total, mean	71.2 $\pm$ 8.8
SIRSI pass $>55\%$ mean, %	72.2
Postop. SIRSI score for RTP, mean	73.9 $\pm$ 2.7
Postop. SIRSI score for did not RTP, mean	41.5
Follow-up duration for postop. SIRSI score, mo	21 $\pm$ 23.7

SIRSI, Shoulder Instability–Return to Sport after Injury; RTP, return to play.

with a mean RTP of 5.3 months (SD 1.6). Preoperative SIRSI scores had a mean of 55, whereas postoperative SIRSI scores had a mean of 86.5 (SD 11.5). Average follow-up duration for postoperative SIRSI score collection was 8.0 months.

Discussion

This systematic review established that SIRSI scores can be used as a predictive survey when determining RTP in the population undergoing open Latarjet surgery. Hurley et al¹⁸ was one of the first to correlate SIRSI scores with RTP following open Latarjet, reporting that 80% of their cohort returned to play with an average SIRSI score of 70.2. With the set pass rate of the SIRSI questionnaire being ≥ 55 , this is a predictive tool for determining an athlete's psychological readiness after Latarjet.¹² Furthermore, with an average postoperative SIRSI score of 41.5 for those not returning to play following Latarjet, this passing score

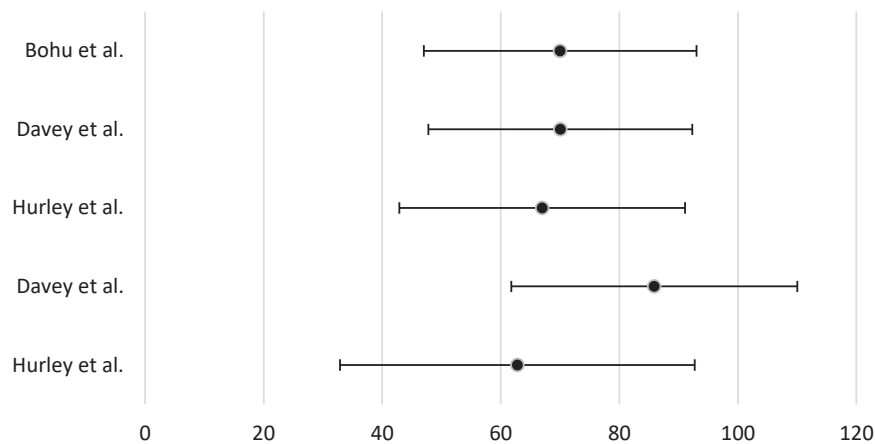


Figure 2 SIRS score by study. *SIRS*, Shoulder Instability–Return to Sport after Injury.

Table IV Reasons why athletes did not return to play

	RTP rate, %	Did not RTP, n	Reasons why did not RTP
Hurley et al ¹⁸	80	8	4 shoulder injury; 4 lifestyle reasons
Hurley et al ¹⁹	93.5	4	3 shoulder injury; 1 lifestyle reasons
Paul et al ²⁴	58	8	5 shoulder injury
Davey et al ⁸	89.7	10	5 shoulder injury; 5 other
Rossi et al ²⁷	94	4	2 not feel psychologically confident; 2 feared reinjury

RTP, return to play.

reaffirms the predictiveness of the SIRS score for RTP. Bohu et al³ also demonstrated this by showing an improvement of SIRS scores from 51 preoperatively to an average score of 70 postoperatively and a 73% RTP.

With the increasing popularity of the Latarjet surgery to effectively manage anterior shoulder instability, it is important to understand the factors regarding RTP outside of physical readiness.²⁵ The most commonly reported reason why athletes did not RTP in our study was “shoulder injury.” Shoulder injury can be interpreted in multiple manners and should be considered multifactorial. One subset of this complex that should be given importance is pain. Like most reconstructive surgeries, postoperative pain is a possible complication. Tanaka et al³⁰ found that when comparing open Latarjet to open Bristow for anterior shoulder instability in rugby players, those who underwent open Latarjet had significantly higher pain postoperatively. This can be due to multiple factors

pertaining to the Latarjet procedure. Godenèche et al¹³ found that 14 of 21 patients who had persistent pain following open Latarjet experienced complete relief following screw removal.

Following shoulder injury, fear of reinjury was the second-most common reason why athletes did not RTP in our review. Nwachukwu et al found that the most commonly reported barrier from RTP was the fear or reinjury.²³ There can also be some speculation that those citing “shoulder injury” as their reason to not RTP were in fact fearful from their previous shoulder injury and did not want to risk recurrence after already going through Latarjet surgery and rehabilitation. Of note, most studies that have analyzed psychological factors for RTP after athlete injury mainly pertain to anterior cruciate ligament reconstruction. Although it is understandable that these are 2 different conditions, 2 different reconstruction methods, and different mechanisms of injury, the fear of reinjury is a ubiquitous reason that is seen across all aspects of RTP after sports injury. Not being psychologically ready was also reported in the studies analyzed. Although this was seen at the same rate as fear of reinjury in our study pool, the literature reports it as a less frequent reason for not returning to play, such as Chen et al⁶ only seeing 1.9% of patients in their systematic review not being psychologically ready compared with 5.8% that feared reinjury.

Today’s athletes expect to return to their sport in a timely and optimal manner. A previous study found this expectation from the athlete’s perspective was as high as 95.5% in those ranging from 18–78 years old.³² Although there are surveys that have been validated regarding the physical demand that a sport requires, such as Roberts and Funk²⁶ for shoulder injuries in rugby players, the SIRS score should be considered just as meaningful with the emphasis being placed on psychological factors for RTP. Even though the SIRS score may seem general and simplistic, it has managed to predict with accuracy the psychological readiness of those who underwent open

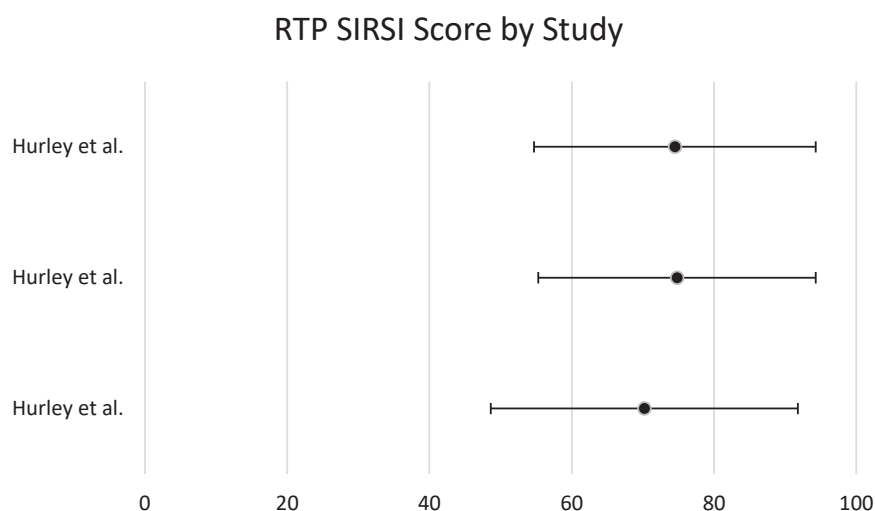


Figure 3 SIRSI score by study for athletes who returned to play. *SIRSI*, Shoulder Instability–Return to Sport after Injury.

Latarjet and returned to play. Surgeons may want to consider using the SIRSI score when determining whether to allow an athlete to RTP, and if the athlete scores poorly, should implement a program to help prepare the athlete for competition. SIRSI scores should also be trended and therefore gathered at the preintervention and post-intervention levels to determine an athlete's rate of improvement as well as net amelioration. This is no different than looking at strength measurements in a return to sport test after an anterior cruciate ligament reconstruction and developing a program to optimize quadriceps strength when an athlete shows deficiency. Further work is needed to better understand psychological readiness around returning to sport.

Limitations

Our review is not without limitations. Sample sizes of the studies were relatively small, increasing the risk of a type II statistical error, where the potential of failing to reject the null hypothesis when it is false exists. Like many systematic reviews, there is heterogeneity of results when reporting outcomes, restricting our ability to pool results together for stronger analyses, such as differentiating by the type and level of sport. Some studies reported SIRSI scores for the entire cohort as an average, whereas other studies reported SIRSI scores for only the cohort that was able to RTP. This resulted in analyzing articles in groups depending on the method they reported SIRSI scores. Another limitation of our study is the athletic level of patients described across all studies that were reviewed. Patients were often not classified exclusively as contact or noncontact, and level of competition varied across included studies as well. This did not allow us to cohesively analyze all groups and resulted in the exclusion of some studies

from various subanalyses. However, despite these limitations, this systematic review of the available data related to RTP using the SIRSI scores supports the importance of understanding the psychological aspects of return to sport after open Latarjet surgery.

Conclusion

Athletes who RTP have higher average SIRSI scores than those who are unable to RTP. Of the athletes who documented why they did not RTP, more than half cited a shoulder injury as their reason for not returning to play, while fear of reinjury and lack of psychological readiness were other common reasons.

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Supplementary Data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jse.2023.11.006>.

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